

Limits of Accuracy

Worksheet 2 (Calculator)

1. W16/qp21/q1(d)

A program showed an athlete running 100 meters, measured correct to the nearest meter. The time the athlete took was 11.3 seconds, measured correct to the nearest 0.1 second. Calculate the upper bound of the athlete's average speed. [2]

2. W18/qp21/q9(b)

The mass of lamp C is 340g, correct to the nearest 10g. 8 of these lamps are placed in a packing case. The total mass of the packing case and the 8 lamps is 4.2kg, correct to the nearest 0.1kg. Calculate the upper bound of the mass of the packing case when empty. Give your answer in kilograms. [3]

3. W20/qp22/q4(b)

A different solid is in the shape of a cuboid. The cuboid measures 8cm by 4cm by 6 cm. These measurements are given correct to the nearest centimeter. Calculate the lower bound of the volume of the cuboid. [2]

4. s22/qp21/q3(c)

The upper bound for the area of a rectangular piece of fabric is $8.8125m^2$. The width of the piece of fabric is 2.3 meters, correct to the nearest 0.1m. The length of the piece of fabric is d meters, correct to the nearest 0.1m. Find the value of d. [3]

5. s23/qp22/q6(c)

Sophia cycles a distance of 3000m correct to the nearest 10 meters. She cycles this distance in a time of 450 seconds correct to the nearest 10 seconds. Calculate the upper bound for her average cycling speed in meters per second. [3]

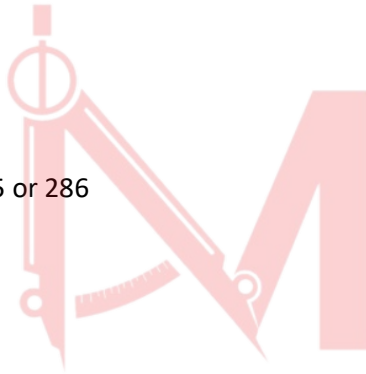
6. s23/qp41/q3(c) 0580

On a farm, there are 500 hens, correct to the nearest 10.

- i. In one year, the mean number of eggs laid per hen was 320 eggs, correct to the nearest 20. Calculate the upper bound for the total number of eggs all the hens lay in that year. [3]
- ii. Another farm has 800 hens, correct to the nearest 20. Calculate the lower bound for the difference between the number of hens on the two farms. [2]

Answer Key - Worksheet 2

1. 17
2. 1.57
3. 144.375
4. 3.7
5. 6.75
6. (i) 166 650 or 165816 (ii) 285 or 286



Mathlete^o
= by Saad